Pre-conception Health: Homework Help

Year 10 Child Development · Topic 1.1: Factors affecting pre-conception health for women and men

How to use this sheet

Read each **explanation** first so you understand it, then write the answer on your sheet **in your own words**. Your teacher wants to see that *you* understand it, and putting it in your own words is how it sticks in your memory for the exam. Then try the quiz on the Child Development page.

Task 1: Why does pre-conception health matter?

a) What is meant by the term "pre-conception health"?

What this really means

Break the word up. **Pre** means *before*. **Conception** is the moment a sperm fertilises an egg and a baby starts to grow. So **pre-conception health** is how healthy the mum **and** dad are **before** the baby is made. It is about their bodies (weight, diet, illnesses) and their lifestyle choices (smoking, drinking, drugs).

Model answer

Pre-conception health is the physical and mental health of **both the man and the woman** in the months **before they try to conceive** a baby. It includes their lifestyle choices, such as diet, weight, exercise, smoking, alcohol and drug use, and steps like taking **folic acid** and checking vaccinations, to give them the best chance of getting pregnant and having a healthy baby.

b) Why is good pre-conception health important? (4 marks)

What this really means

It is worth **4 marks**, so give **4 separate reasons**. Each one needs a short **"because"** explaining it. The big idea is that the baby's body starts forming **very early**, often before anyone knows a baby is on the way. And sperm take **about 2 to 3 months** to grow, so the dad's health months earlier matters too.

Model answer

1. It **increases fertility**, so the couple are more likely to conceive, because healthy eggs and sperm are more likely to meet and fertilise.
2. It **reduces the risk of miscarriage, premature birth and low birth weight**, because the mother's body is in the best condition to support a pregnancy.
3. It **reduces the risk of birth defects**, because the baby's brain, spine and heart form in the first few weeks, often **before the woman knows she is pregnant**. For example, taking **folic acid** lowers the risk of **spina bifida**.
4. The **man's health matters too**, because sperm take about **3 months** to develop, so smoking, alcohol or drugs in that time can damage the sperm that make the baby.

Exam tip

Mentioning the **man** as well as the woman is an easy way to pick up marks, because lots of students forget about him.

Task 2: Lifestyle factors and fertility (10 marks)

There are 5 factors and 2 columns, so 10 boxes and probably **1 mark per box**. For each box, say **what happens in the body** and **how that makes conceiving harder**. Key words are in **bold**. Try to use them in your answers.

Factor	Effect on women	Effect on men
Weight **Remember:** Weight affects **hormones**. Hormones control eggs and sperm.	Being **overweight** or **underweight** can upset the hormones that control **ovulation** (releasing an egg each month), so periods can become irregular and it is harder to get pregnant. Obesity also raises the risk of **miscarriage** and **gestational diabetes**. A healthy BMI is 18.5 to 24.9.	Being overweight can **lower testosterone** and reduce the **number and quality of sperm**, so there is less chance of a sperm reaching and fertilising the egg.
Smoking **Remember:** Smoking damages **both** the egg and the sperm. Second-hand smoke counts too.	The chemicals in cigarettes **damage eggs** and the fallopian tubes, so it can take longer to conceive. It raises the risk of **miscarriage** and **ectopic pregnancy** (the baby growing outside the womb) and can bring on the menopause earlier.	Smoking **lowers sperm count**, makes sperm swim less well (poor **motility**) and can **damage the DNA** inside sperm. It can also cause erection problems.
Drinking alcohol **Remember:** For women the safest amount is **none**. For men, keep it low.	Alcohol can **disrupt the menstrual cycle** and reduce fertility. A woman may not know she is pregnant for several weeks, and alcohol in that time can harm the baby and cause **Fetal Alcohol Spectrum Disorder (FASD)**. The NHS advice is **no alcohol** when trying for a baby.	Heavy drinking can **lower testosterone**, reduce **sperm quality and count** and cause impotence (problems having sex). Men are advised to stay within **14 units a week**.
Taking recreational drugs **Remember:** Drugs can **switch off** egg release and sperm production.	Drugs such as cannabis and cocaine can **stop or disrupt ovulation**. If she becomes pregnant, drugs pass to the baby and can cause **low birth weight**, **premature birth** and **withdrawal symptoms** in the newborn.	Cannabis can **reduce sperm count and quality**. **Anabolic steroids** (used for bodybuilding) can stop sperm production altogether. Cocaine also damages sperm.
Parental age **Remember:** Age is the **only** factor here you **cannot change**.	A woman is born with all her eggs, and they get **fewer and lower quality** with age. Fertility starts to fall in the early 30s and **faster after 35**. Older mothers have a higher risk of miscarriage and of conditions such as **Down's syndrome**. Very young mothers (teenagers) have higher risks too, such as premature birth.	Men make new sperm all their lives, so fertility falls **more slowly**. From about **40**, sperm quality drops, it can take longer to conceive, and there is a slightly higher risk of miscarriage and of some conditions in the baby.

Key words explained

Fertility = being able to get pregnant (women) or get someone pregnant (men).
Ovulation = when an ovary releases an egg, about once a month.
Sperm count = how many sperm there are. **Motility** = how well they swim.
Miscarriage = losing a baby before 24 weeks of pregnancy.
Ectopic pregnancy = the baby starts growing outside the womb, usually in a fallopian tube. It is dangerous and the pregnancy cannot continue.

Task 3: Ranking activity

Which factor has the greatest effect on a couple's ability to conceive?

What this really means

This is an **opinion** question, so there is **no single right answer**. Your teacher is marking your **reasons**, not just the order. To decide, ask three questions about each factor:

- Does it affect **both** the man and the woman, or just one?
- Does it **stop** fertility completely, or just **lower** it?
- How **common** is it?

The sheet lists 4 factors in the ranking table. Age is not in it, probably because it is the one factor nobody can change.

Here is **one sensible ranking** with reasons. You can disagree, as long as you explain why.

Rank	Factor	Why it goes here
1	**Smoking**	It harms **both** partners at once (eggs **and** sperm), it is very common, and it also raises the risk of miscarriage and ectopic pregnancy.
2	**Taking recreational drugs**	Some drugs can **stop** ovulation or sperm production completely (for example steroids), which is a bigger effect than just lowering it.
3	**Weight**	It affects hormones in both partners and is very common, but the effect builds up gradually and can be improved by changing diet and exercise.
4	**Drinking alcohol**	Heavy drinking harms fertility, but light drinking has a smaller effect on *conceiving*. Its biggest danger is to the baby once pregnant.

How to write your answer

Write one sentence for your number 1, like this:

*"I think **smoking** has the greatest effect because it damages **both** the woman's eggs and the man's sperm, so it lowers the fertility of both partners at the same time."*

Sources: NHS (nhs.uk: "Trying to get pregnant" and "Infertility"), Tommy's (tommys.org: "Planning for pregnancy"), OCR Cambridge Nationals Child Development J809 specification.